E-commerce KPI Report

UK online gift retailer (wholesale and retail customers) · Source: Online Retail II, UCI ML Repository (CC BY 4.0) · Sample project built with public data

Report period Dec 2010 to Nov 2011 **Compared with** Dec 2009 to Nov 2010 Months 12

Choose the first and last month in the yellow cells. Every KPI, ranking and chart below updates automatically.

Net revenue	Orders	Average order value	Active customers	Repeat customer rate
£9,330,919	18,957	£508.14	4,293	63.9%
▲ 1.9 % vs comparison	▼ 4.0 % vs comparison	▲ 6.8 % vs comparison	▲ 1.3 % vs comparison	▼ 2.5 pp vs comparison

KPI summary

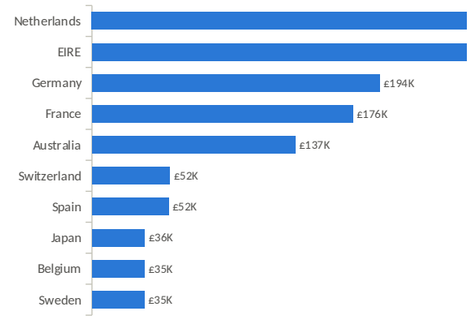
KPI	Current period	Comparison period	Change
Net revenue	£9,330,919	£9,156,117	+1.9%
Gross sales	£9,632,728	£9,396,642	+2.5%
Returns	-£301,810	-£240,525	+25.5%
Return rate (% of gross sales)	3.1%	2.6%	+0.6 pp
Orders	18,957	19,743	-4.0%
Average order value	£508.14	£475.95	+6.8%
Units sold (net)	5,062,258	5,428,938	-6.8%
Active customers	4,293	4,239	+1.3%
Repeat customers (2+ orders)	2,742	2,813	-2.5%
Repeat customer rate	63.9%	66.4%	-2.5 pp
Revenue from repeat customers	93.1%	93.9%	-0.8 pp
Revenue from guest checkouts	15.0%	11.6%	+3.4 pp
UK share of net revenue	84.5%	86.1%	-1.7 pp

Returns are negative values. 'pp' = percentage points. Definitions: Notes sheet.

Top 10 international markets (excluding the UK)

#	Country	Net revenue	Orders	Share of total
1	Netherlands	£271,752	90	2.9%
2	EIRE	£252,257	272	2.7%
3	Germany	£193,775	426	2.1%
4	France	£175,685	365	1.9%
5	Australia	£136,923	56	1.5%
6	Switzerland	£52,483	50	0.6%
7	Spain	£51,503	86	0.6%
8	Japan	£35,537	19	0.4%
9	Belgium	£35,383	94	0.4%
10	Sweden	£35,166	33	0.4%

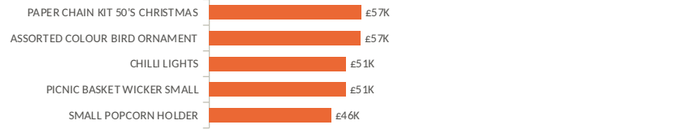
Top international markets (selected period)



Top 10 products by net revenue

#	Stock code	Description	Units	Net revenue	Share of total
1	22423	REGENCY CAKESTAND 3 TIER	12,556	£158,569	1.7%
2	85123A	WHITE HANGING HEART T-LIGHT HOLDER	34,517	£97,408	1.0%
3	47566	PARTY BUNTING	17,870	£97,325	1.0%
4	85099B	JUMBO BAG RED RETROSPOT	46,166	£89,980	1.0%
5	23084	RABBIT NIGHT LIGHT	26,340	£57,058	0.6%
6	22086	PAPER CHAIN KIT 50'S CHRISTMAS	16,788	£56,847	0.6%
7	84879	ASSORTED COLOUR BIRD ORNAMENT	34,952	£56,630	0.6%
8	79321	CHILLI LIGHTS	9,761	£51,113	0.5%
9	22502	PICNIC BASKET WICKER SMALL	1,862	£51,024	0.5%
10	22197	SMALL POPCORN HOLDER	50,229	£45,558	0.5%

Top 10 products (selected period)



Monthly trend (all months; the selected period is highlighted)

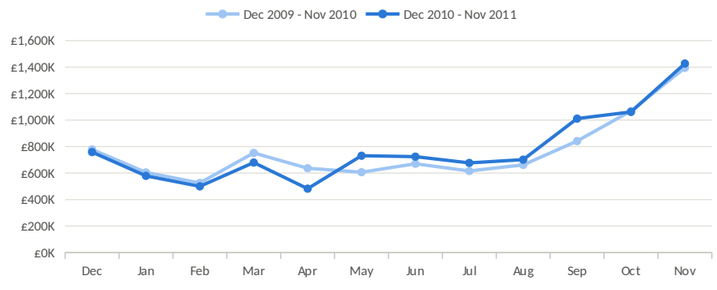
Month	Gross sales	Returns	Net revenue	Orders	Avg order value	New customers*	Change vs previous month	Change vs same month last year
Dec 2009	£797,901	-£19,558	£778,343	1,666	£478.93	951	n/a	n/a
Jan 2010	£612,297	-£7,466	£604,832	1,049	£583.70	368	-22.3%	n/a
Feb 2010	£537,910	-£12,575	£525,334	1,189	£452.41	375	-13.1%	n/a
Mar 2010	£761,706	-£9,604	£752,102	1,647	£462.48	441	+43.2%	n/a
Apr 2010	£646,341	-£10,025	£636,316	1,435	£450.41	294	-15.4%	n/a
May 2010	£643,492	-£37,193	£606,299	1,484	£433.62	255	-4.7%	n/a
Jun 2010	£696,583	-£26,337	£670,246	1,612	£432.12	267	+10.5%	n/a
Jul 2010	£632,994	-£17,133	£615,862	1,507	£420.04	185	-8.1%	n/a
Aug 2010	£674,038	-£12,398	£661,640	1,402	£480.77	163	+7.4%	n/a
Sep 2010	£869,243	-£27,686	£841,557	1,789	£485.88	239	+27.2%	n/a
Oct 2010	£1,094,436	-£25,451	£1,068,985	2,244	£487.72	375	+27.0%	n/a
Nov 2010	£1,429,701	-£35,099	£1,394,602	2,719	£525.82	326	+30.5%	n/a
Dec 2010	£775,638	-£17,548	£758,091	1,550	£500.41	76	-45.6%	-2.6%
Jan 2011	£670,380	-£91,525	£578,855	1,081	£620.15	72	-23.6%	-4.3%
Feb 2011	£507,800	-£8,336	£499,464	1,093	£464.59	125	-13.7%	-4.9%
Mar 2011	£689,834	-£10,504	£679,329	1,440	£479.05	179	+36.0%	-9.7%
Apr 2011	£515,378	-£33,316	£482,062	1,235	£417.31	106	-29.0%	-24.2%
May 2011	£739,953	-£8,948	£731,005	1,668	£443.62	111	+51.6%	+20.6%
Jun 2011	£737,559	-£13,714	£723,845	1,525	£483.65	108	-1.0%	+8.0%
Jul 2011	£688,219	-£11,407	£676,812	1,452	£473.98	101	-6.5%	+9.9%
Aug 2011	£724,217	-£22,927	£701,290	1,339	£540.86	107	+3.6%	+6.0%
Sep 2011	£1,028,320	-£16,981	£1,011,340	1,818	£565.63	188	+44.2%	+20.2%
Oct 2011	£1,103,314	-£41,613	£1,061,701	2,005	£550.28	221	+5.0%	-0.7%
Nov 2011	£1,452,116	-£24,991	£1,427,125	2,751	£527.85	191	+34.4%	+2.3%

** New customers = identified customers whose first order in the data falls in that month. Dec 2009 counts every customer active that month, because earlier history is not in the dataset.*

Same months, two years

Month	Dec 09 - Nov 10	Dec 10 - Nov 11
Dec	£778,343	£758,091
Jan	£604,832	£578,855
Feb	£525,334	£499,464
Mar	£752,102	£679,329
Apr	£636,316	£482,062
May	£606,299	£731,005
Jun	£670,246	£723,845
Jul	£615,862	£676,812
Aug	£661,640	£701,290
Sep	£841,557	£1,011,340
Oct	£1,068,985	£1,061,701
Nov	£1,394,602	£1,427,125

Monthly net revenue: this year vs last year



Notes: KPI definitions, data and method

Sample project built with public data. Prepared by Marlon Jiménez Valencia.

KPI definitions

KPI	Definition	How it is calculated
Net revenue	Sales minus returns, in GBP, for invoices dated in the period.	SUMIFS on tbl_Orders[NetRevenue] (returns are negative).
Gross sales	Value of sales invoices only (quantity × unit price).	SUMIFS with Type = "Sale".
Returns	Value of cancelled / returned items (shown as a negative number).	SUMIFS with Type = "Return" (invoice numbers starting with C).
Return rate	Returns as a share of gross sales.	-Returns / Gross sales.
Orders	Number of sales invoices.	COUNTIFS with Type = "Sale".
Average order value (AOV)	Average value of a sales invoice.	Gross sales / Orders.
Units sold (net)	Units sold minus units returned.	SUMIFS on tbl_Orders[Units].
Active customers	Identified customers with at least one sales invoice in the period.	COUNTIF on tbl_Customers[OrdersCur] > 0. Guest checkouts are excluded.
Repeat customers	Active customers with two or more orders in the period.	COUNTIF on tbl_Customers[OrdersCur] > 1.
Repeat customer rate	Share of active customers who ordered more than once.	Repeat customers / Active customers.
Revenue from repeat customers	Share of identified-customer revenue that comes from repeat customers.	SUMIFS on tbl_Customers[RevenueCur] where OrdersCur > 1, divided by total.
Revenue from guest checkouts	Share of net revenue from invoices without a customer ID.	SUMIFS with CustomerID = "Guest" / Net revenue.
New customers	Identified customers whose first order in the data is in that month.	MINIFS gives each customer's first order date; COUNTIFS counts them by month.
Top markets / products	Ranking by net revenue in the selected period.	LARGE on a ranking helper column, then INDEX/MATCH to fetch the name and values.
Comparison period	The same number of months right before the report period.	EDATE(StartMonth, -NMonths). Shown as n/a if it starts before Dec 2009.

Data source

Dataset	Online Retail II: all transactions of a UK-based online retailer of gift-ware, Dec 2009 - Dec 2011. Many customers are wholesalers.
Publisher	UCI Machine Learning Repository (Daqing Chen, 2012). DOI: 10.24432/C5CG6D
Licence	Creative Commons Attribution 4.0 International (CC BY 4.0).
Period used	24 full months: Dec 2009 - Nov 2011 (Dec 2011 has only 9 days and is excluded).
Currency	GBP (pounds sterling). Values are invoice values: quantity × unit price.

Cleaning steps (from the raw file to the Data sheet)

Rule	Note	Rows removed / rows after
Load both sheets (2009-2010 and 2010-2011)	Raw invoice lines	0 removed / 1,067,371 rows
Remove exact duplicate lines	Includes 1-9 Dec 2010, which appears in both sheets	34,335 removed / 1,033,036 rows
Keep 24 full months (Dec 2009 - Nov 2011)	Dec 2011 has only 9 days of data	25,247 removed / 1,007,789 rows
Remove non-product codes	Postage, fees, adjustments, tests and gift vouchers (ADJUST, ADJUST2, AMAZONFEE, B, BANK CHARGES, C2, CRUK, D, DOT, M, PADS, POST, S, TEST001, TEST002, m, gift_*)	5,724 removed / 1,002,065 rows
Remove lines with price <= 0	Free items and stock write-offs	5,911 removed / 996,154 rows
Check: sales lines with quantity <= 0	Validation only: none left after the price rule	0 removed / 996,154 rows
Upper-case stock codes	3,148 lines / 173 codes such as 85123a -> 85123A (Excel lookups ignore case, so variants would be double-counted)	0 removed / 996,154 rows
Standardise text fields	One description per stock code (623 codes had several); missing customer IDs labelled 'Guest'	0 removed / 996,154 rows

Invoice lines are then grouped into 45,968 invoices (Data sheet).

Limitations

- About a fifth of invoice lines have no customer ID ('Guest'). They count for revenue and orders, but not for customer KPIs.
- Returns are counted in the month they were processed, which may differ from the month of the original sale.
- First-order dates only see history from Dec 2009, so 'new customers' are overstated in the first months.
- This is a sample analysis of a public dataset, not a client engagement.